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by Naveen Kumar

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Robust Multi-Robot Mapless Navigation in Dynamic Hospital Environments with Accurate Intention Inference, Trajectory Prediction, and Explainable AI

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01. Introduction

Hospital navigation is challenging due to dynamic pedestrians, uncertain behaviors, and sensor noise. We propose a human-aware multi-modal framework combining perception, human intention prediction, and PPO-based navigation for safe and socially compliant robot movement.

02. Objective

- Human-aware navigation in hospital environments
- Multi-modal perception using LiDAR, Depth, IMU, and Wheel Encoder
- Human tracking and trajectory prediction
- Pedestrian intention estimation
- Sensor fusion for scene representation
- PPO-based socially compliant navigation
- Safe and collision-free robot motion

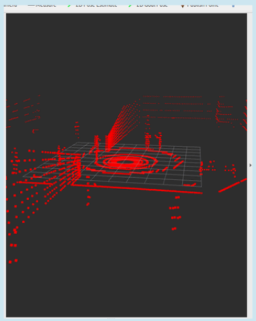
03. Simulation setup

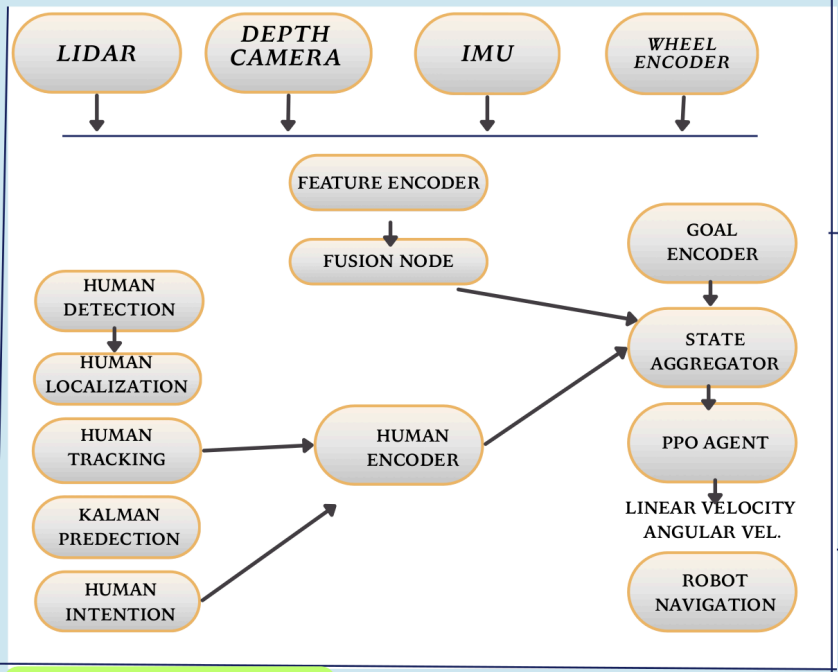
1. **Environment**: Smart Hospital (Gazebo)
2. **Robot**: TurtleBot3
3. **Sensors**: 3D LiDAR, Depth Camera, IMU, Wheel Encoder
4. **Framework**: ROS2 Jazzy, Gazebo, PyTorch, Python



04. LiDAR Encoder

Input: 3D Point Cloud (1024 × 3)
Backbone: PointNet++
Feature Extraction:
SA1: Local geometric features
SA2: Hierarchical feature learning
SA3: Global scene representation
Projection Head:
1024→256→128→64
Output:
lidar∈R64





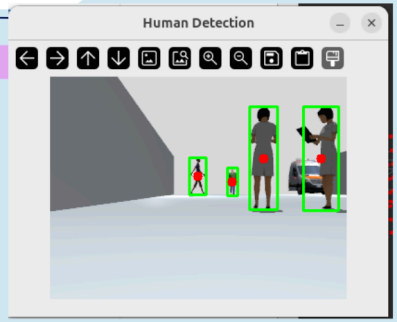
05. Human Detection

Input: RGB Image
Method: Deep Learning-based Human Detector
Output: Human Centroids and Confidence Score
D={u,v,c}

06. Kalman Trajectory Prediction

Input: Human Tracks (x,y)
State Vector:
X=[x,y,vx,vy]
Prediction:
xf = x + vxT
yf = y + vyT

Output: Future Human Position



07. Human Intention Prediction

- Uses speed and heading change
- Classes: STOP, STRAIGHT, LEFT, RIGHT
- θ=atan2(dy,dx)
- Output: Intention Label

08. State Aggregator

- Environment Feature (128D)
- Human Feature (30D)
- Goal Feature (2D)

S=[Fenv,Fhuman,Fgoal]
S∈R160S
Output: 160-D Navigation State for PPO

09. PPO Navigation

1. State: 160D
2. Action: (v,ω)
3. Policy:
$\pi\theta(a|s)$
4. Rewards
   - Goal Achievement
   - Collision Avoidance
   - Social Compliance
   - Smooth Navigation

Output: Safe and Human-Aware Robot Motion

10. Conclusion

- Human-aware multi-modal navigation framework developed.
- Integrated perception, prediction, and intention modeling.
- PPO enables safe and socially compliant navigation.
- Suitable for dynamic hospital environments.

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